

# Assignment 10: Data Scraping

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## OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

## Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

## Set up

1. Set up your session:
  - Load the packages `tidyverse`, `rvest`, and any others you end up using.
  - Check your working directory

```
#1
library(tidyverse);library(lubridate);library(viridis);library(here)
library(rvest);library(dataRetrieval);library(tidycensus);library(zoo)
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
  - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
  - Scroll down and select the LWSP link next to Durham Municipality.
  - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

#2

```
URL <- "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024"
website <- read_html(URL)
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- PWSID
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

#3

```
water_system <- website %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()
PWSID <- website %>%
  html_nodes("td tr:nth-child(1) td:nth-child(5)") %>%
  html_text()
Owner <- website %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()
MGD <- website %>%
  html_nodes("th~ td+ td") %>%
  html_text()
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It’s likely you won’t be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: “Jan”, “May”, “Sept”, “Feb”, etc...

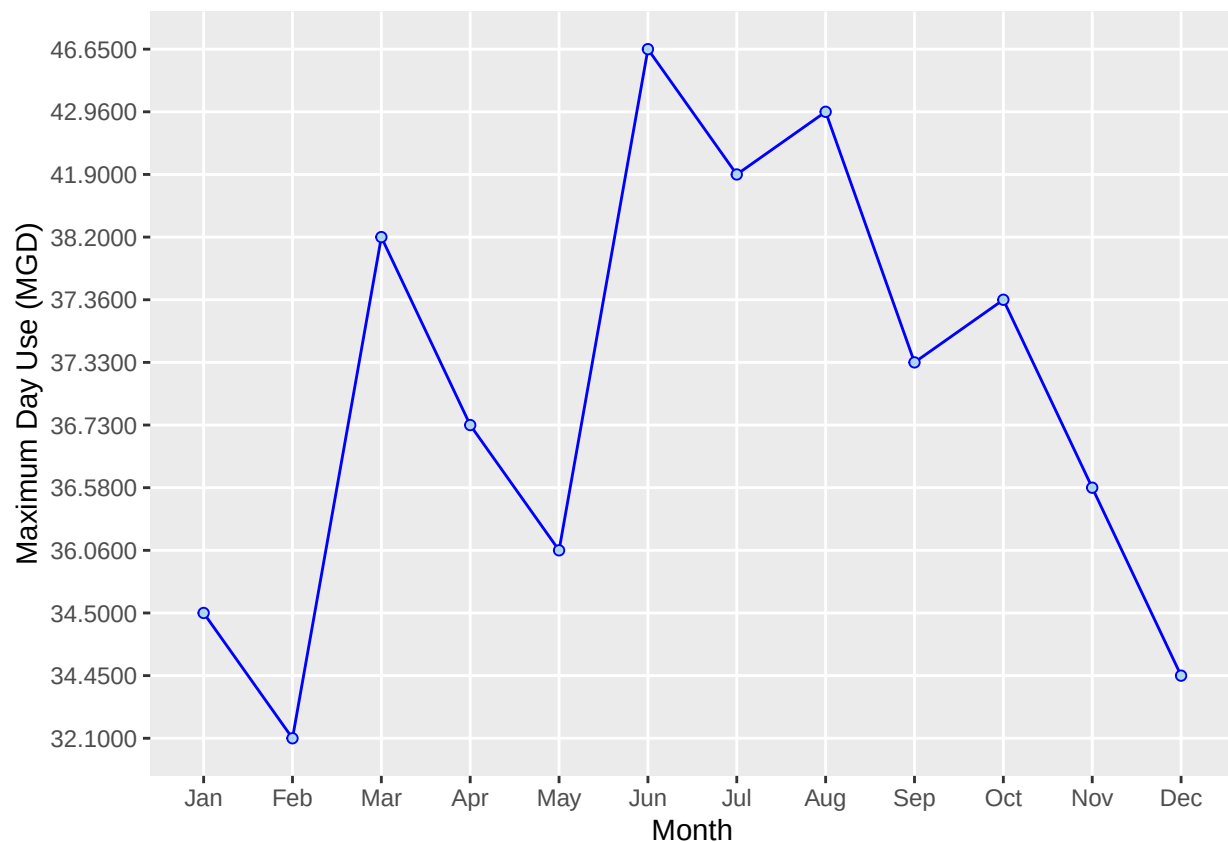
5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

#4

```
df <- data.frame("Water System" = water_system,
  "Owner" = Owner,
  "PWSID" = PWSID,
  "Month" = c(1,5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12),
  "Max Day Use (MGD)" = MGD,
  "Year" = rep(2024)) %>%
mutate(Month = factor(month.abb[Month],
  levels = month.abb,
  ordered = TRUE),
  Date = as.yearmon(paste(Year, Month), "%Y %b"))
```

#5

```
ggplot(df, aes(x = Month, y = Max.Day.Use..MGD., group = 1)) +
  geom_line(color = "blue") +
  geom_point(shape = 21, color = "blue", fill = "lightblue") +
  labs(x = "Month", y = "Maximum Day Use (MGD)")
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - "PWSID" and "year" - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)

- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

```
#6.
scrape.it <- function(Year, PWSID){

  #Retrieve the website contents
  website <- read_html(paste0('https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=',
                              PWSID, '&year=', Year))

  water_system <- website %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>% html_text()
  Owner <- website %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>% html_text()
  MGD <- website %>%
  html_nodes("th~ td+ td") %>% html_text()

  #Convert to a dataframe
  df_collect <- data.frame("WaterSystem" = water_system,
                           "Owner" = Owner,
                           "PWSID" = PWSID,
                           "Month" = c(1,5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12),
                           "MaxDayUse" = MGD,
                           "Year" = rep(Year, 12)) %>%
  mutate(Month = factor(month.abb[Month],
                        levels = month.abb,
                        ordered = TRUE),
         Date = as.yearmon(paste(Year, Month), "%Y %b"))

  #Pause for a moment - scraping etiquette
  #Sys.sleep(1) #uncomment this if you are doing bulk scraping!

  #Return the dataframe
  return(df_collect)
}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7

Durham <- scrape.it(2020, '03-32-010')
glimpse(Durham)

## Rows: 12
## Columns: 7
## $ WaterSystem <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "Durham"~
## $ Owner <chr> "Municipality", "Municipality", "Municipality", "Municipal~
## $ PWSID <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-32~
```

```
## $ Month      <ord> Jan, May, Sep, Feb, Jun, Oct, Mar, Jul, Nov, Apr, Aug, Dec
## $ MaxDayUse  <chr> "36.0100", "36.9800", "41.6900", "32.0500", "40.6100", "40~
## $ Year       <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020~
## $ Date       <yearmon> Jan 2020, May 2020, Sep 2020, Feb 2020, Jun 2020, Oct 2020~
```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

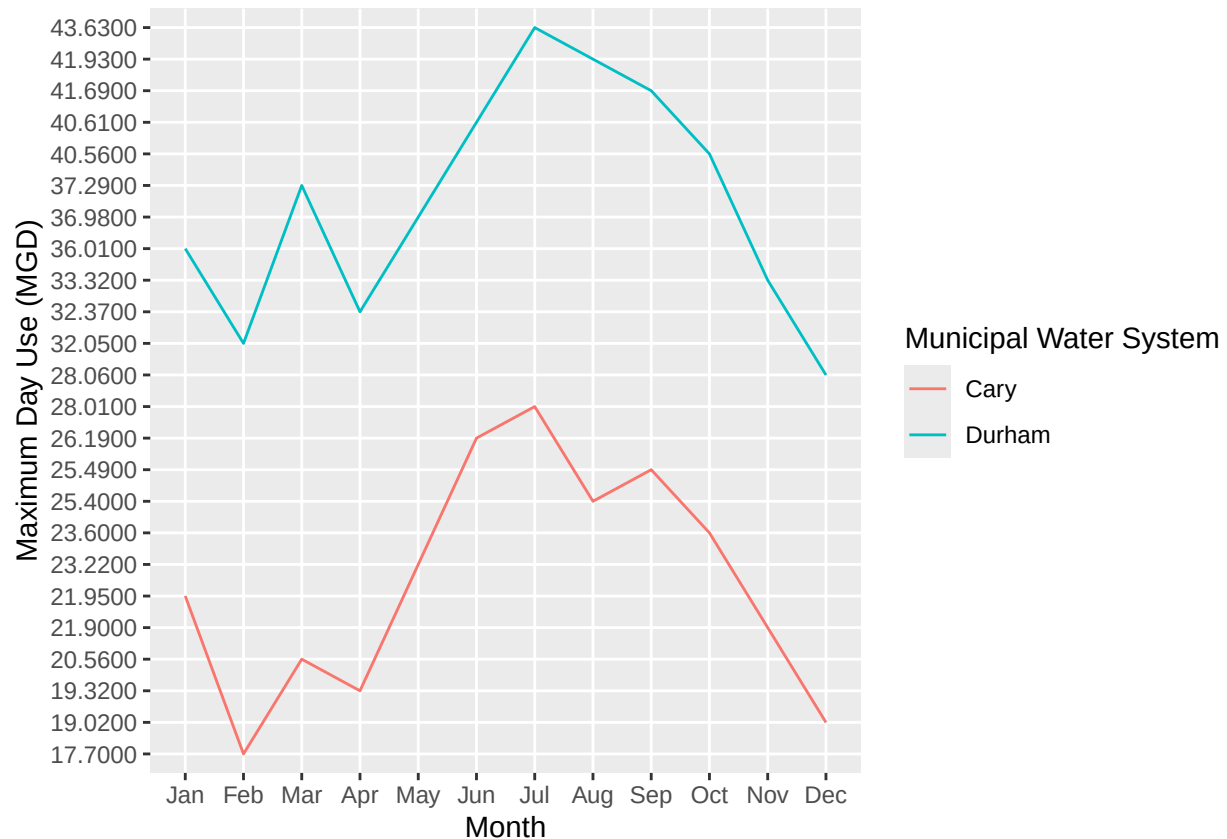
#8

```
Cary <- scrape.it(2020, '03-92-020')
glimpse(Cary)
```

```
## Rows: 12
## Columns: 7
## $ WaterSystem <chr> "Cary", "Cary", "Cary", "Cary", "Cary", "Cary", "Cary", "C~
## $ Owner       <chr> "Municipality", "Municipality", "Municipality", "Municipal~
## $ PWSID       <chr> "03-92-020", "03-92-020", "03-92-020", "03-92-020", "03-92~
## $ Month       <ord> Jan, May, Sep, Feb, Jun, Oct, Mar, Jul, Nov, Apr, Aug, Dec
## $ MaxDayUse   <chr> "21.9500", "23.2200", "25.4900", "17.7000", "26.1900", "23~
## $ Year        <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020~
## $ Date        <yearmon> Jan 2020, May 2020, Sep 2020, Feb 2020, Jun 2020, Oct 2020~
```

```
combined_water <- bind_rows(Durham, Cary)

ggplot(combined_water, aes(x = Month, y = MaxDayUse, color = factor(WaterSystem),
                           group = WaterSystem)) +
  geom_line() +
  labs(x = "Month", y = "Maximum Day Use (MGD)", color = "Municipal Water System")
```



9. Use the code & function you created above to plot Cary’s max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = ‘loess’).

TIP: See Section 3.2 in the “10\_Data\_Scraping.Rmd” where we apply “map2()” to iteratively run a function over two inputs. Pipe the output of the map2() function to bind\_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
Years <- rep(2018:2023)

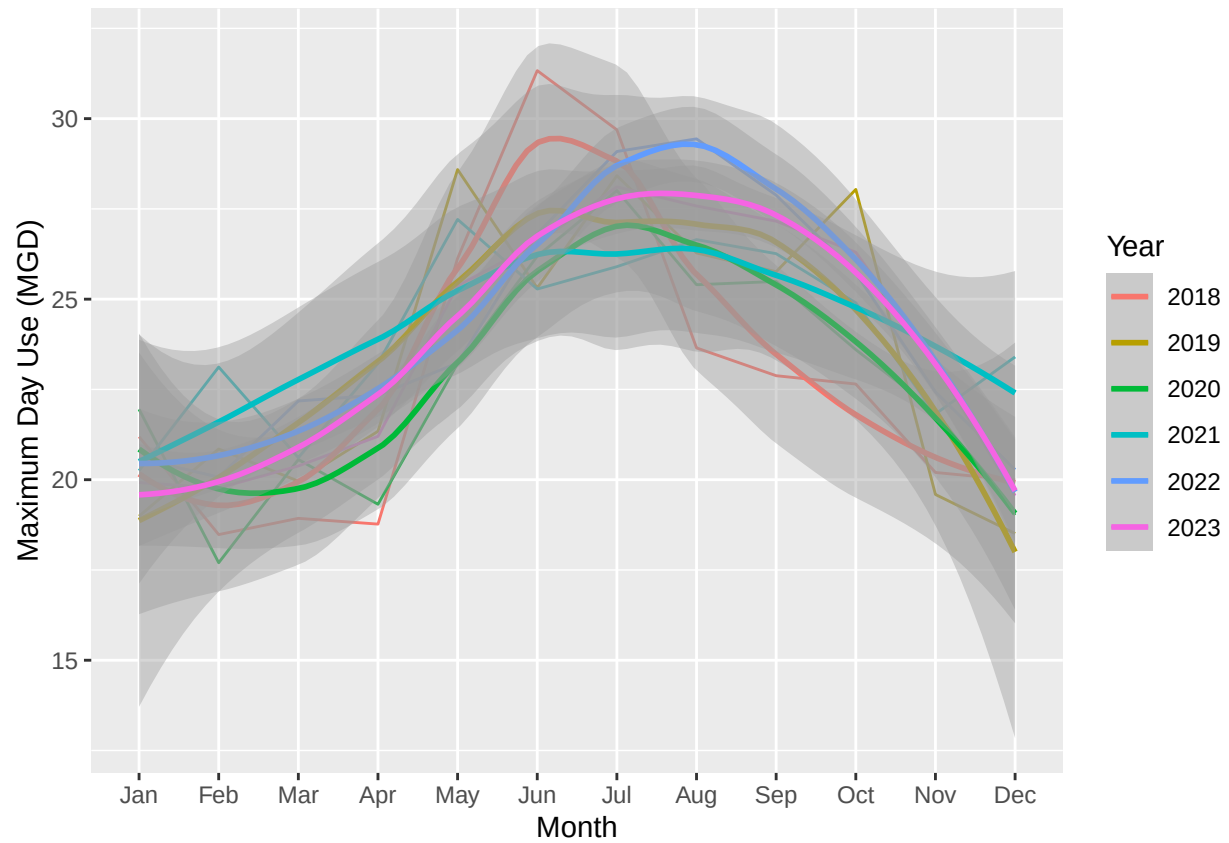
# "Map" the "scrape.it" function to retrieve data for all these
dfs_Cary <- map2(Years, '03-92-020', scrape.it)

# Conflate the returned list of dataframes into a single one
df_Cary <- bind_rows(dfs_Cary)

df_Cary$MaxDayUse <- as.numeric(df_Cary$MaxDayUse)

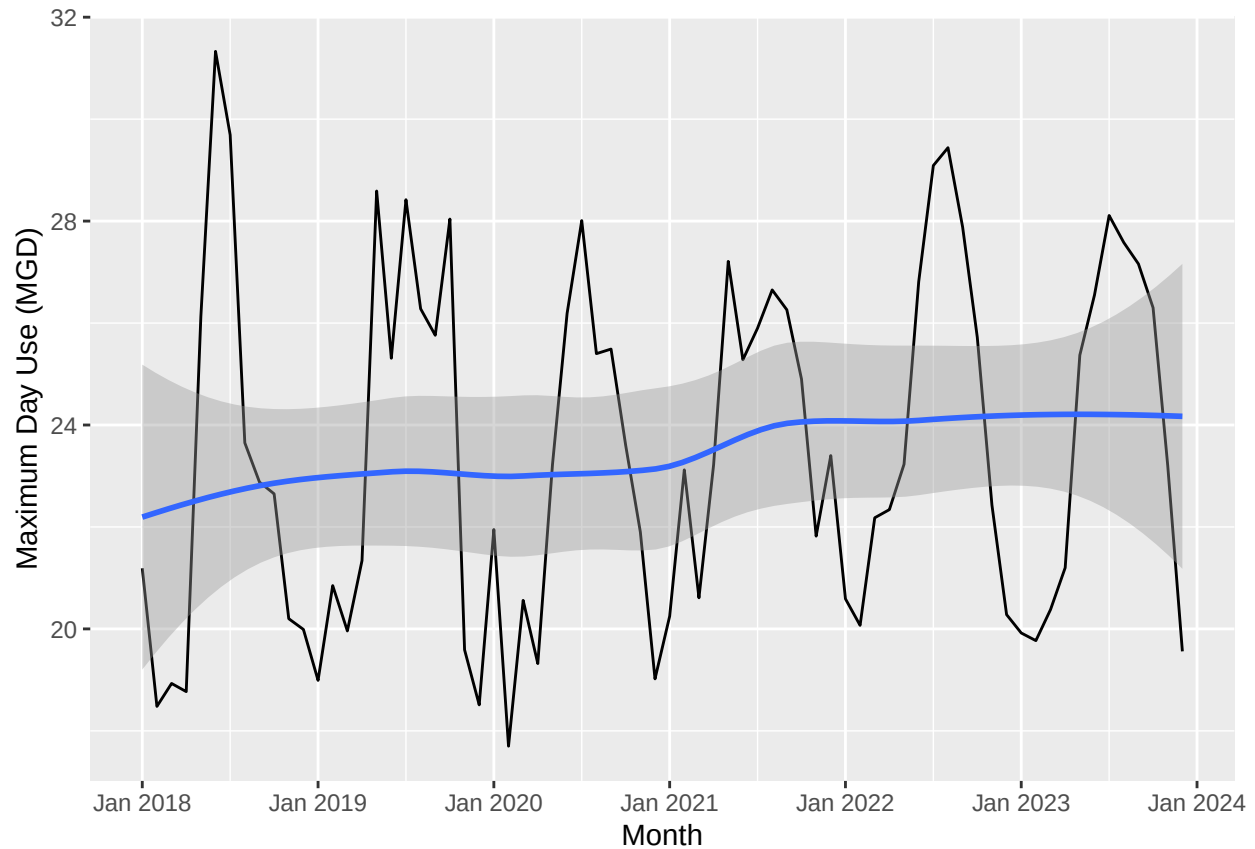
ggplot(df_Cary, aes(x = Month, y = MaxDayUse, color = factor(Year), group = Year)) +
  geom_line() +
  geom_smooth(method = 'loess') +
  labs(x = "Month", y = "Maximum Day Use (MGD)", color = "Year")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
ggplot(df_Cary, aes(x = Date, y = MaxDayUse)) +
  geom_line() +
  geom_smooth(method = 'loess') +
  labs(x = "Month", y = "Maximum Day Use (MGD)")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Based on the graphs, Cary uses significantly more water in the summer months than the winter months, but the magnitude of useage in the summer does not appear to have a trend. However, it looks like slightly more water has been used in winter months in more recent years. >